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Thermal oxidization systems and methods with greenhouse gas capture

Abstract

A thermal oxidizer employing an oxidation mixer, an oxidation chamber, a retention chamber and a heat dissipater forming a fluid flow path for thermal oxidation of a waste gas. In operation, the oxidation mixer facilitates a combustible mixture of the waste gas and an oxidant into an combustible waste gas stream, the oxidation chamber facilitates a primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream, the retention chamber facilitates a secondary combustion reaction of the oxygenated waste gas stream into oxidized gases and the heat dissipater reduces the temperature of the flow of oxidized gases within the heat dissipater, which is communicated to a greenhouse gas processor that extracts greenhouse gas(es) from the vaporized oxidized gases. The greenhouse gas processor may condensate the greenhouse gas(es), acid neutralize the condensation of the greenhouse gas(es), and capture the acid neutralization of the condensation of the greenhouse gas(es).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation-in-part of U.S. application Ser. No. 16/313,368 entitled “Thermal Oxidization Systems and Methods,” filed Dec. 26, 2018, which is a national stage entry of PCT/US17/39575 entitled “Thermal Oxidization Systems and Methods,” filed Jun. 27, 2017, which claims priority to U.S. Provisional Application No. 62/354,991 entitled “Thermal Oxidization Systems and Methods,” filed Jun. 27, 2016, (collectively the “Priority Applications”). This present application claims priority and benefit of the Priority Applications. The content of the Priority Applications is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present disclosure generally relates to a thermal oxidation of waste gas (e.g., toxic release inventory (“TRI”) gases, particularly volatile organic compound (“VOC”)) into desirable oxidized gases including carbon dioxide gas (CO.sub.2), water vapor (H.sub.2O), nitrogen gas (N.sub.2) and oxygen gas (O.sub.2).

(2) The present disclosure specifically relates to a controlled thermal oxidation of the waste gas involving a regulated combustible mixture of the waste gas and an oxidant (e.g., atmospheric air), a regulated combustion reaction of the combustible mixture into desirable oxidized gases and/or a regulated atmospheric venting of desirable oxidized gases.

BACKGROUND OF THE INVENTION

(3) FIG. 1 illustrates a thermal oxidizer **20** known in the art of the present disclosure. Thermal oxidizer **20** employs a heating chamber **21** and a perforated oxidation reaction (“POR”) chamber **24** forming a fluid flow path for a desired oxidation of a waste gas stream **32** (e.g., TRI gases) into oxidized gases **41** (e.g., CO.sub.2 , H.sub.2O , N.sub.2 and O.sub.2).

(4) In operation, a heating element **22** within heating chamber **21** is activated to generate heat waves **23** for heating waste gas stream **32** as waste gas stream **32** is feed from a waste gas source **30** via an inlet conduit **31** through heating chamber **21** into POR chamber **24** and as an oxidant **10** (e.g., atmospheric air) flows into POR chamber **24**. The heating of waste gas stream **32** is intended to facilitate a combustible oxidation **40** of waste gas stream **32** within POR chamber **24** into oxidized gases **41**. However, there are design flaws with thermal oxidizer **20** that impedes a combustible oxidation **40** of waste gas stream **32** within POR chamber **24** into oxidized gases **41**.

(5) First, thermal oxidizer **20** fails to regulate a combustible mixture of waste gas stream **32** and oxidant **10** within a flammable range between an upper explosive limit (“UEL”) and a lower explosive level (“LEL”) as known in the art of the present disclosure, particularly when waste gas stream **32** enters heating chamber **21** at a concentration below the LEL for waste gas stream **32** (i.e., waste gas is too lean). Specifically, a simultaneous suction of oxidant **10** into POR chamber **24** and venting of any gases within POR chamber **24** impedes a sufficient flow of oxidant **10** into POR chamber **24** to ensure a combustible mixture of waste gas stream **32** and oxidant **10** within the flammable range. Consequently, while the combustible mixture of waste gas stream **32** and oxidant **10** may be within the flammable range upon a power-on of thermal oxidizer **20**, the mixture of waste gas stream **32** and oxidant **10** will eventually become too “rich” for combustible oxidation **40** within POR chamber **24**. As a result, an incomplete combustion facilitates a buildup of high concentrations of undesirable gases (e.g., carbon monoxide (CO)) and soot within POR chamber **24**.

(6) Second, thermal oxidizer **20** fails to regulate a combustion reaction of waste gas stream **32** within POR chamber **24**. Specifically, thermal oxidizer **20** does not regulate the heating of heating element **22**. Consequently, additional undesirable gases (e.g., nitrogen oxides (NO) and (NO.sub.2)) (not shown) may form within POR chamber **24** if a temperature of heating waves **23** is too high. Moreover, even if the heating of heating element **22** was regulated, POR chamber **24** is configured and sized for an instantaneous combustible oxidation **40** of waste gas stream **32** without a sufficient retention time for a combustible mixture of waste gas stream **32** and oxidant **10**, if any, to convert to desirable oxidized gases **41** (e.g., CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2).

(7) The inventions of the present disclosure overcome the drawbacks of prior heating element based thermal oxidizers, particularly thermal oxidizer **20** of FIG. **1**.

SUMMARY OF THE INVENTION

(8) One form of the inventions of the present disclosure is a thermal oxidizer employing an oxidation mixer, an oxidation chamber, a retention chamber and a heat dissipater forming a fluid flow path for thermal oxidation of a waste gas (e.g., toxic release inventory ("TRI") gases, particularly volatile organic compound ("VOC")).

(9) In operation, the oxidation mixer facilitates a combustible mixture of the waste gas and an oxidant (e.g., atmospheric air) into a combustible waste gas stream.

(10) For purposes of the inventions of the present disclosure, the term "combustible waste gas stream" broadly encompasses any stream of gas including molecules combinable with oxygen or combust resulting in heat and light, and excludes totally inert gases.

(11) The oxidation mixer may be any form of any oxidation mixer as known in the art of the present disclosure and hereinafter conceived including, but not limited to, a venturi or distributed air-gas mixer.

(12) The oxidation mixer may be fed the waste gas via any type of waste gas feeding mechanism known in the art of the present disclosure and hereinafter conceived including, but not limited to, (1) an aeration nozzle, (2) an aeration nozzle and back flow preventer and (3) an aeration nozzle, back flow preventer and blocking valve.

(13) The oxidation mixer may be fed the oxidant via any type of oxidant feeding mechanism as known in the art of the present disclosure and hereinafter conceived including, but not limited to, (1) an open air inlet involving an inductive air/waste gas flow optionally providing flash back protection, (2) a force modulation air blower with a mixing "T" or (3) a forced modulation air pump and mixing "T".

(14) A heating element or a gas burner of the oxidation chamber generate heat waves facilitating a primary combustion reaction of the combustible waste gas stream flowing from the oxidation mixer to the oxidation chamber.

(15) For purposes of the inventions of the present disclosure,

(16) (1) the term "heating element" broadly encompasses any element for converting electricity into heat through the process of Joule/ohmic/resistive, inductive or other means of electrical heating,

(17) (2) the term "gas burner" broadly encompasses an element for igniting a natural gas stream,

(18) (3) the term "primary combustion reaction" broadly encompasses an oxidation of the combustible waste gas stream flowing from the oxidation mixer into the oxidation chamber involving a partial combustion of the combustible waste gas stream within the oxidation chamber resulting in an oxygenated waste gas stream, and

(19) (4) the term "oxygenated waste gas stream" broadly encompasses a partial combustion of the combustible waste gas stream in many forms including, but not limited to, CO+other hydrocarbon compounds+CO.sub.2+H.sub.2O+N.sub.2+excess O.sub.2.

(20) The oxygenated waste gas stream flows from the oxidation chamber into the retention chamber whereby the retention chamber facilitates a secondary combustion reaction as needed of the oxygenated waste gas stream into oxidized gases. For purposes of the inventions of the present

disclosure, the term “secondary combustion reaction” encompasses a complete oxidation of the oxygenated waste gas stream flowing from the oxidation chamber to the retention chamber involving a conversion of the oxygenated waste gas stream into oxidized gases (e.g., CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2). The retention chamber may also include an additional heating element to facilitate the complete oxidation of oxygenated waste gas stream into oxidized gases. (21) Concurrently or alternatively, the oxidation chamber may further include an supplemental oxidant inlet for mixing additional oxidant to the oxygenated waste gas stream into a combustible oxygenated waste gas stream flowing into the retention chamber. (22) The heat dissipater facilitates an atmospheric venting of the oxidized gases flowing into the heat dissipater from the retention chamber. (23) Another form of inventions of the present disclosure is a thermal oxidization system including a thermal oxidizer of the present disclosure and a greenhouse gas processor. For purposes of the inventions of the present disclosure, the term “greenhouse gas processor” broadly encompasses any machine structurally configured to extract greenhouse gas(es) within oxidized gases. (24) For this thermal oxidization system, a heat dissipator of the thermal oxidizer heat dissipates the flow of oxidized gases within the heat dissipator and communicates the oxidized gases at a reduced temperature to the greenhouse gas processor. (25) In operation, the greenhouse gas processor extracts greenhouse gas(es) from within the oxidized gases, and may further vent non-greenhouse gas(es) within the oxidized gases. (26) In extracting the greenhouse gas(es) from within the oxidized gases, the greenhouse gas processor may liquid condensate the greenhouse gas(es), acid neutralize the condensation of the greenhouse gas(es) and capture the acid neutralized condensation of the greenhouse gas(es). (27) The foregoing forms and other forms of the present invention as well as various features and advantages of the present invention will become further apparent from the following detailed description of various embodiments of the present invention read in conjunction with the accompanying drawings. The detailed description and drawings are merely illustrative of the present invention rather than limiting, the scope of the present invention being defined by the appended claims and equivalents thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a block diagram of an exemplary thermal oxidizer as known in the art.
- (2) FIG. 2 illustrates a block diagram of a first exemplary chamber embodiment of a thermal oxidization system in accordance with the inventive principles of the present disclosure.
- (3) FIG. 3 illustrates a block diagram of an exemplary embodiment of a control system in accordance with the inventive principles of the present disclosure.
- (4) FIGS. 4A-4C illustrate a flowchart representative of an exemplary embodiment of a thermal oxidization method in accordance with the inventive principles of the present disclosure.
- (5) FIGS. 5A and 5B illustrate perspective views of a first exemplary embodiment of the thermal oxidization system of FIG. 2 in accordance with the inventive principles of the present disclosure.
- (6) FIGS. 6A-6C illustrate exploded views of the thermal oxidization system of FIGS. 5A and 5B.
- (7) FIG. 7 illustrates a perspective view of a second exemplary embodiment of the thermal oxidization system of FIG. 2 in accordance with the inventive principles of the present disclosure.
- (8) FIG. 8 illustrates a block diagram of a second exemplary embodiment of a thermal oxidization system in accordance with the inventive principles of the present disclosure.
- (9) FIG. 9 illustrates a perspective view of an exemplary embodiment of the thermal oxidizer of FIG. 8 in accordance with the inventive principles of the present disclosure.
- (10) FIG. 10A illustrates an exemplary atmospheric embodiment of the thermal oxidizer of FIG. 9.

- (11) FIG. 10B illustrates an exemplary forced air embodiment of the thermal oxidizer of FIG. 9.
- (12) FIG. 11 illustrates a first exemplary embodiment of a gas fired thermal oxidizer in accordance with the inventive principles of the present disclosure.
- (13) FIG. 12 illustrates a flowchart representative of an exemplary embodiment of a gas fired thermal oxidization method in accordance with the inventive principles of the present disclosure.
- (14) FIG. 13A illustrates an exemplary ignition phase of the flowchart of FIG. 12.
- (15) FIG. 13B illustrates an exemplary combustion phase of the flowchart of FIG. 12.
- (16) FIG. 14 illustrates a second exemplary embodiment of a gas fired thermal oxidizer in accordance with the inventive principles of the present disclosure.
- (17) FIG. 15 illustrates an exemplary embodiment of greenhouse gas processor based thermal oxidation system of the present disclosure in accordance with the inventive principles of the present disclosure.
- (18) FIG. 16 illustrates an exemplary embodiment of the greenhouse gas processor based thermal oxidation system of FIG. 15 in accordance with the inventive principles of the present disclosure.
- (19) FIG. 17 illustrates a flowchart representative of an exemplary embodiment of greenhouse gas processing method in accordance with the inventive principles of the present disclosure.
- (20) FIG. 18 illustrates an exemplary embodiment of the vaporizing heat dissipator of FIG. 17 in accordance with the inventive principles of the present disclosure.
- (21) FIG. 19 illustrates exemplary embodiments of a condenser and an acid gas neutralizer of FIG. 1 in accordance with the inventive principles of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (22) To facilitate an understanding of inventions of the present disclosure, the following description of FIGS. 2-4 teach basic inventive principles of thermal oxidization systems and thermal oxidization methods of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of thermal oxidization systems and thermal oxidization methods of the present disclosure.
- (23) Referring to FIG. 2, a thermal oxidization system of the present disclosure incorporates a thermal oxidizer 50a employing an oxidation mixer 51, an oxidation chamber 52a, a retention chamber 53a and a heat dissipater 54a forming a fluid flow path for oxidation of a waste gas stream 32 (e.g., TRI gases) into oxidized combustion products 74 (e.g., oxidized gases 73 including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2).
- (24) Oxidation mixer 51 is structurally configured for implementing a combustible mixture of an oxidant 10 (e.g., atmospheric air) supplied by oxidant supply 11 (e.g., a blower or a pump) via a supply line 12 and of a waste gas stream 32 (e.g., TRI gases) supplied by a waste gas source 30 as controlled via a control flow conduit 33 (e.g., a solenoid valve and a flame arrestor in series coupling waste gas source 30 to oxidation mixer 51) into a combustible waste gas stream 70.
- (25) In one embodiment, oxidation mixer 51 is a venturi air-gas mixer whereby turbulent fluid flows of oxidant 10 and waste gas stream 32 into the venture air-gas mixer are controlled via oxidant supply 11 and control flow conduit 33 to ensure combustible waste gas stream 70 attains proportional concentrations of oxidant 10 and waste gas stream 32 within a flammable range (e.g., 11.5:1 oxidant 10 to waste gas stream 32). Additionally, oxidation mixer 51 may be equipped with a nozzle (not shown) for regulating a feeding of combustible waste gas stream 70 into oxidation chamber 52a whereby the nozzle may be structurally configured to generate more turbulence to combustible waste gas stream 70.
- (26) Oxidation chamber 52a is structurally configured for implementing a primary combustion reaction 71 therein of combustible waste gas stream 70 into an oxygenated waste gas stream 72 via a controlled emission of heat waves 56 by a heating element 55. In one embodiment, oxidation chamber 52a is a refractory ceramic cylinder and heating element 55 is embedded within the walls of the refractory ceramic cylinder.

(27) Optionally, oxidation chamber **52a** may further employ a spark igniter **66** for a controlled ignition of combustible waste gas stream **70** at a proximal opening of oxidation chamber **52a**. For this embodiment, if oxidization mixer **51** is equipped with a nozzle, then a distal tip of spark igniter **66** may be positioned within or adjacent to the flow of the combustible waste gas stream **70** out of the nozzle into oxidation chamber **52a**.

(28) Retention chamber **53a** is structurally configured for implementing a retention time for a secondary combustion reaction of oxygenated waste gas stream **72** into heated oxidized combustion products **74** (e.g., oxidized gases **73** including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2). In one embodiment, retention chamber **53a** is a refractory ceramic cylinder integrated with oxidation chamber **52a** as shown.

(29) Heat dissipater **54a** is structurally configured for implementing a heat exchange of atmosphere air **75** with heated oxidized combustion products **74** to vent cooled oxidized combustion products **74** into the atmosphere. In one embodiment, heat dissipater **54a** includes a heat exchanger **57a** constructed of stainless steel woven fabric, which has been pleated and rolled into a cylinder shape whereby oxidized combustion products **74** exits heat exchanger **57a** along a length and circumference of the vertical wall of heat exchanger **57a** as shown with cooling atmosphere air **75** being directed vertically past the vertical wall of heat exchanger **57a** to thereby extract heat from heat dissipater **54a**.

(30) Additionally, heat dissipater **54a** may be equipped with mesh baffles (e.g., mesh baffles **58b** and **58c** as shown) axially aligned on a longitudinal axis of the cylindrical heat exchanger **57a** to thereby provide a more controlled flow diversion of oxidized combustion products **74** in a direction of vertical wall of heat exchanger **57a** as shown in FIG. 2.

(31) Referring to FIG. 3, thermal oxidization system of the present disclosure further incorporates a control system employing an oxidation controller **61**, a data logger **63** and a data reporter **65** housed within a control box **60**.

(32) Oxidation controller **61** is structurally configured for controlling an operation of thermal oxidizer **50a** as will be further described herein in connection with a description of FIGS. 4A-4C.

(33) In one embodiment, oxidation controller **61** is an application specific main board or an application specific integrated circuit for controlling a thermal oxidation application of various inventive principles of the present disclosure as subsequently described herein in connection with FIGS. 4A-4C. The structural configuration of oxidation controller **61** may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for controlling an execution of the thermal oxidation application.

(34) A non-limiting example of oxidation controller **61** is an all-in-one XL4 controller sold by Horner APG of Indianapolis, Ind. that is configured in accordance with the inventive principles of the present disclosure.

(35) Data logger **63** is structurally configured for logging operational data ("OD") **62** transmitted by oxidation controller **61** to data logger **63** via a push or pull operation, or by a monitoring of specific data points of oxidation controller **61** by data logger **63**. Operational data **62** includes data informative of an operational status of thermal oxidizer **50a** in executing the oxidation of waste gas stream **32**.

(36) In one embodiment, data logger **63** is an application specific main board or an application specific integrated circuit for controlling a data logging application of the present disclosure. The structural configuration of data logger **63** may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware

stored on non-transitory computer readable medium(s)) for executing the data logging application.
(37) A non-limiting example of data logger **63** is an all-in-one XLE controller sold by Horner APG of Indianapolis, Ind. that is configured in accordance with the inventive principles of the present disclosure.

(38) Another non-limiting example of data logger **63** is as an application module configured within oxidation controller **61**.

(39) Also in practice, data logger **63** may be omitted and oxidation controller **61** may be configured for executing the data logging application of the present disclosure.

(40) Data reporter **65** is structurally configured for receiving reporting data (“RD”) **64** from data logger **63** via a push or pull operation and transmitting reporting data **64** to a network **80** (e.g., a cellular network). Reporting data **64** is operational data **62** processed by data logger **63** into a reporting format associated with data reporter **65**.

(41) In one embodiment, data reporter **65** is an application specific modem for executing a data reporting application of the present disclosure.

(42) A non-limiting example of data logger **66** is a SonicU™ modem sold by SonicU, LLC of Greenfield, Ind. that is configured in accordance with the inventive principles of the present disclosure.

(43) Referring back to FIG. 2, oxidation controller **61** regulates a control of thermal oxidizer **50a** in accordance with a thermal oxidization method of the present disclosure as represented by a flowchart **90** shown in FIGS. 4A-4C. In support of executing the thermal oxidization method: 1. oxidation controller **61** senses and controls a flash point temperature T.sub.FP of oxidation chamber **52a** via a connection of thermocouple **67** at a signal input **67s**; 2. oxidation controller **61** senses and controls an oxidation point temperature T.sub.Op of retention chamber **53a** via a connection of thermocouple **68** at a signal input **68s**; 3. oxidation controller **61** senses and controls a venting point temperature T.sub.FP of heat dissipater **54a** via a connection to thermocouple **69** at a signal input **69s**; 4. oxidation controller **61** controls a feeding rate of oxidant **10** by oxidant supply **11** via a connection of a signal output **13s** to oxidant supply **11a**; 5. oxidation controller **61** controls a feeding rate of waste gas stream **32** by waste gas source **30** via a connection of a signal output **34s** to control flow conduit **33**; 6. oxidation controller **61** controls an activation and a deactivation of heating element **55** via a signal output **55s**; and 7. oxidation controller **61** controls an activation and a deactivation of spark igniter **66** via a signal output **66s**;

(44) Referring to FIGS. 2 and 4A, prior to be oxidation controller **61** being powered-on, oxidant supply **11** is deactivated and control flow conduit **33** is closed.

(45) Upon oxidation controller **61** being powered-on, oxidation controller **61** executes flowchart **90**, and initiates a transmission of a point monitoring of flash point temperature T.sub.FP, oxidization point temperature T.sub.Op and venting point temperature T.sub.VP to data logger **63** for recording purposes.

(46) As to the execution of flowchart **90** by oxidation controller **61**, a stage **S92** of flowchart **90** encompasses a thermocouple validity check (“TVC”) by oxidation controller **61** concurrently or sequentially involving:

(47) 1. a comparison by oxidation controller **61** of flash point temperature T.sub.FP relative to a flash thermocouple validity threshold (“FTVT”);

(48) 2. a comparison by oxidation controller **61** of oxidation point temperature T.sub.OP relative to an oxidation thermocouple validity threshold (“OTVT”); and

(49) 3. a comparison by oxidation controller **61** of venting point temperature T.sub.VP relative to a venting thermocouple validity threshold (“VTVT”).

(50) In practice, to test the validity of the thermocouples, the thermocouple validity thresholds should be set at an anticipated maximum room temperature (e.g., 22° C.) or to an anticipated maximum local outdoor air temperature.

(51) For a TVC cycle of stage **S92**, if flash point temperature T.sub.FP is more than flash

thermocouple validity threshold FTVT, OR if oxidation point temperature T.sub.OP is more than oxidation thermocouple validity threshold OTVT, OR if venting point temperature T.sub.VP is more than venting thermocouple validity threshold VTVT, then oxidation controller **61** is powered down for a maintenance inspection of the thermocouple(s) and powered on again after a successful repair/replacement, if any, of the thermocouple(s).

(52) Otherwise, if flash point temperature T.sub.FP is less than or equal to flash thermocouple validity threshold FTVT, AND if oxidation point temperature T.sub.OP is less than or equal to oxidation thermocouple validity threshold OTVT, AND if venting point temperature T.sub.VP is less than or equal to venting thermocouple validity threshold VTVT, then oxidation controller **61** proceeds to a stage **S94** of flowchart **90** encompassing a thermal oxidation limit check (“TOLC”) by oxidation controller **61** concurrently or sequentially involving:

(53) 1. a comparison by oxidation controller **61** of flash point temperature T.sub.FP relative to a flash thermal limit threshold (“FTLT”);

(54) 2. a comparison by oxidation controller **61** of oxidation point temperature T.sub.OP relative to an oxidation thermal limit threshold (“OTLT”); and

(55) 3. a comparison by oxidation controller **61** of venting point temperature T.sub.VP relative to a venting thermal limit threshold (“VTLT”).

(56) In practice, to ensure a safe heat activation of oxidation chamber **52a**, the thermal limit thresholds should represent a predefined maximum temperature of oxidation chamber **52a**, retention chamber **53a** and heat dissipater **54a** prior to the heat activation of thermal oxidizer **50a** (e.g., 300° F. for all thermocouple limit thresholds).

(57) For a TOLC cycle of stage **S94**, if flash point temperature T.sub.FP is more than flash thermal limit threshold FFLT, OR if oxidation point temperature T.sub.OP is more than oxidation thermal limit threshold OTLT, OR if venting point temperature T.sub.VP is more than venting thermal limit threshold VTLT, then oxidation controller **61** returns to stage **S94** to execute a new TOLC cycle.

(58) Otherwise, if flash point temperature T.sub.FP is less than or equal to flash thermal limit threshold FFLT, AND if oxidation point temperature T.sub.OP is less than or equal to oxidation thermal limit threshold OTLT, AND if venting point temperature T.sub.VP is less than or equal to venting thermal limit threshold VTLT, then oxidation controller **61** proceeds to a stage **S96** of flowchart **90** to execute a pre-oxidation stage of the thermal oxidation of waste gas stream **32** sequentially involving:

(59) 1. with control flow conduit **33** being closed via valve control signal VC (e.g., a binary open-close signal) and heating element **55** being deactivated via heat regulation signal HR (e.g., an AC current), oxidation controller **61** controlling a gaseous purge of thermal oxidizer **50a** by activating oxidant supply **11** via a supply control signal SC (e.g., a digital amplitude signal) for a specified time period. For example, oxidation controller **61** may activate oxidant supply **11** via supply control signal for an oxidation feed of oxidant **10** (e.g., atmospheric air) into oxidation mixer **51** at a 100% oxidation feed rate of oxidized gas **10** for thirty (30) seconds to one (1) minute (e.g., the feed rate in cubic feet/minute over a duration exceeding at least four (4) times a volume of oxidation chamber **52a** and retention chamber **53a**);

(60) 2. after completion of the purge, oxidation controller **61** controlling an activation of oxidant supply **11** via supply control signal SC for an oxidation feed of oxidant **10** into oxidation mixer **51** at a 50% oxidation feed rate; and

(61) 3. concurrent with or subsequent to the 50% oxidation feed rate of oxidant **10** into oxidation mixer **51**, oxidation controller **61** executing a zero crossover regulation of heating element **55** via heating regulation signal HR.

(62) Upon executing stage **S96**, oxidation controller **61** proceeds to a stage **S98** of flowchart **90** encompassing an oxidation chamber ignition check (“OCIC”) by oxidation controller **61** involving a comparison by oxidation controller **61** of flash point temperature T.sub.FP relative to a oxidation chamber ignition threshold (“OCIT”) (e.g., 1100° F.).

(63) Oxidation controller **61** cycles through stage **S98** until such time flash point temperature T.sub.FP is equal to or greater than oxidation chamber ignition threshold OCIT whereby oxidation controller **61** proceeds to a stage **S100** of flowchart **90** to execute an oxidation stage of the thermal oxidation of waste gas stream **32** sequentially involving:

(64) 1. if spark ignitor **57** is employed, oxidation controller **61** controlling a spark ignition of combustible waste gas stream **70** within oxidation chamber **52a** by activating spark igniter **57** via a spark ignition signal **S1** provided to a power supply of spark igniter **57** (not shown for clarity)(e.g., a spark plug transformer based power circuit). The spark ignition is executed to facilitate the primary combustion reaction **71** of combustible waste gas stream **70** within oxidation chamber **52a**; and

(65) 2. oxidation controller **61** controlling an activation of a waste gas feed (e.g., 0.35 cubic feet/minute; 21 cubic feet/hour) into oxidation mixer **51** by opening control flow conduit **33** via valve control signal **VC** to form combustible waste gas stream **70** whereby oxidant **10** flowing into oxidation mixer **51** will create vacuum on waste gas stream **30** at the inlet of the oxidization mixer **51**; and

(66) 3. oxidation controller **61** controlling a regulation of a secondary combustion reaction of oxygenated waste gas stream **72** within retention chamber **53a** by modulating the oxidation feed rate of oxidant **10** by oxidant supply **11** based on a comparison of oxidation point temperature T.sub.OP relative to a oxidation feed modulation threshold (“OFMT”) (e.g., 1400° F.). For example, the oxidation feed rate of oxidant **10** is increased if the oxidation point temperature T.sub.OP is less than the oxidation feed modulation threshold OFMT and conversely, the oxidation feed rate of oxidant **10** is decreased if the oxidation point temperature T.sub.OP is greater than the oxidation feed modulation threshold OFMT.

(67) Upon executing stage **S100**, oxidation controller **61** proceeds to a stage **S102** of flowchart **90** encompassing an oxidation protection check (“OPC”) by oxidation controller **61** concurrently or sequentially involving:

(68) 1. a comparison by oxidation controller **61** of flash point temperature T.sub.FP relative to a flash point protection threshold (“FPPT”);

(69) 2. a comparison by oxidation controller **61** of oxidation point temperature T.sub.OP relative to an oxidation point protection threshold (“OPPT”); and

(70) 3. a comparison by oxidation controller **61** of venting point temperature T.sub.VP relative to a venting point protection threshold (“VPPT”).

(71) In practice, to ensure oxidation protection of thermal oxidizer **50**, the protection thresholds should represent a predefined maximum temperature of oxidation chamber **52a**, retention chamber **53a** and heat dissipater **54a** during oxidation stage **S100** (e.g., 1400° F. for oxidation chamber **52a**, 1800° F. for retention chamber **53a** and 300° F. for heat dissipater **54a**).

(72) For a OPC cycle of stage **S102**, if flash point temperature T.sub.FP is less than flash point protection threshold FPPT, AND if oxidation point temperature T.sub.OP is less than oxidation point protection threshold OPPT, AND if venting point temperature T.sub.VP is less than venting point protection threshold VPPT, then oxidation controller **61** proceeds to a stage **S104** of flowchart **90** to be subsequently described herein.

(73) Otherwise, if flash point temperature T.sub.FP is equal to or greater than flash point protection threshold FPPT, OR if oxidation point temperature T.sub.OP is equal to or greater than oxidation point protection threshold OPPT, OR if venting point temperature T.sub.VP is equal to or greater than venting point protection threshold VPPT, then oxidation controller **61** proceeds to a stage **S106** of flowchart **90** to execute a shutdown stage of the thermal oxidation of waste gas stream **32** concurrently or sequentially involving:

(74) 1. oxidation controller **61** controlling a termination of waste gas feed into oxidation mixer **51** by closing control flow conduit **33** via valve control signal **VC**;

(75) 2. oxidation controller **61** controlling a deactivation of the zero crossover regulation of heating

element **55** via heat regulation signal **HR**; and

(76) 3. oxidation controller **61** controlling an attenuation of the oxidation fee rate of oxidant **10** to zero (0) via supply control signal **SC**.

(77) Upon completion of stage **S106**, oxidation controller **61** executes a power-off routine.

(78) Still referring to FIG. **4C**, stage **S104** encompasses a waste gas limit check (“WGLC”) by oxidization controller **61** involving:

(79) 1. a comparison by oxidation controller **61** of oxidation point temperature **T.sub.OP** relative to a waste gas limit threshold (“WGLT”); and

(80) 2. a comparison by oxidation controller **61** of an open duration **OD.sub.CV52** of control flow conduit **33** relative to a fixed time period **X** in any unit of time (e.g., seconds as shown).

(81) In practice, the waste gas limit threshold represents a predefined temperature indicative of a failure to achieve a primary combustion reaction **71** within oxidation chamber **53a** after time period **X** (e.g., ten (10) seconds) to thereby impede a flow of an unacceptable quantity of waste gas stream **32** through chambers **52a** and **53a**.

(82) For a WGLP cycle of stage **S104**, if oxidation point temperature **T.sub.OP** is equal to or greater than waste gas limit threshold **WGLT** OR if open duration **OD.sub.CV52** of control flow conduit **33** is less than **X** seconds, then oxidation controller **61** returns to stage **S102** as previously described herein.

(83) Otherwise, if oxidation point temperature **T.sub.OP** is less than waste gas limit threshold AND if open duration **OD.sub.CV52** of control flow conduit **33** is equal to or greater than **X** seconds, then oxidation controller **61** proceeds to stage **S106** as previously described herein.

(84) Referring back to FIG. **3**, previously stated herein, upon oxidation controller **61** being powered-on, data logger **62** initiates a recording of flash point temperature **T.sub.FP**, oxidization point temperature **T.sub.OP** and venting point temperature **T.sub.VP** over a course of an execution of flowchart **90** by oxidation controller **61**.

(85) Furthermore, oxidation controller **61** may provide a local reporting of flash point temperature **T.sub.FP**, oxidization point temperature **T.sub.OP** and venting point temperature **T.sub.VP** via a graphical display and/or data reporter **65** may provide a remote reporting of flash point temperature **T.sub.FP**, oxidization point temperature **T.sub.OP** and venting point temperature **T.sub.VP** via a wireless communication to network **80**.

(86) In practice, oxidation controller **61** and/or data logger **63** may be further structurally configured to detect conditions suitable for a warning (e.g., a visual and/or audial alarm, and/or a text message) to be locally issued by oxidation controller **61** or remotely issued by data reporter **65**. Such conditions include, but are not limited to, (1) a continual cycling of stage **S94** indicative of an operational issue with thermal oxidizer **50**, (2) an inability of oxidation chamber **52a** to reach oxidation combustion ignition threshold **OCIT** during stage **S98**, and (3) an occurrence offstage **S104** as an indication of an operational issue of thermal oxidizer **50**.

(87) In practice, the structural dimensions and material compositions of oxidation chamber **52**, retention chamber **53a** and heat dissipater **54a** will be dependent upon the specific oxidation application (e.g., **20,000** BTU/hour of waste gas oxidization).

(88) To facilitate a further understanding of the inventions of the present disclosure, the following description of FIGS. **5-7** teach a couple of embodiments of the thermal oxidation system of FIG. **2**. From this description; those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various additional embodiments of thermal oxidization systems of the present disclosure.

(89) Referring to FIGS. **5A** and **5B**, an embodiment **150a** of thermal oxidizer **50a** (FIG. **2**) employs:

(90) 1. oxidant supply **11** (FIG. **2**) in the form of an air blower **111a**;

(91) 2. oxidation mixer **51** (FIG. **2**) in the form of a venturi air-gas mixer **151a** (not shown in FIGS. **5A** and **5B**; shown in FIG. **6A**);

(92) 3. a control flow conduit **33** in the form of a solenoid valve **133a** equipped with a flame arrestor **133b**;

(93) 4. oxidation chamber **52a** (FIG. 2) in the form an oxidation chamber **152** including a cylindrical insulation blanket and a cylindrical thermal mass defining an oxidation fluid path with a spiral heating element **155** disposed within the oxidation fluid path;

(94) 5. retention chamber **53a** (FIG. 2) in the form of a retention chamber **153** including a cylindrical insulation blanket and a cylindrical thermal mass **154b** defining a retention fluid path;

(95) 6. heat dissipater **54a** (FIG. 2) in the form of a heat dissipater **154**;

(96) 7. a flash thermocouple **67** (FIG. 2) in the form of a flash thermocouple **167**;

(97) 8. an oxidization thermocouple **68** (FIG. 2) in the form of an oxidization thermocouple **168**;

and

(98) 9. a vent thermocouple **69** (FIG. 2) in the form of a vent thermocouple **169**.

(99) Also shown in FIGS. 5A and 5B is a control box **160** as an embodiment of control box **60** (FIG. 3).

(100) FIG. 6A provides a view of venturi air-gas mixer **151a**.

(101) FIG. 6B provides a view of a nozzle **151b** of venturi air-gas mixer **151a**, an interior wall of oxidization chamber **153**, a heating element **155** helically disposed within interior wall of oxidization chamber **153** and an interior wall of retention chamber **154**.

(102) FIG. 6C provides a view of pleated heat exchanger **157** of heat dissipater **154** encircling a top portion of the interior wall of retention chamber **154**.

(103) Referring to FIG. 7, an embodiment **150b** of thermal oxidizer **50a** (FIG. 2) is a modification of thermal oxidizer **150a** (FIG. 5A) involving a substitution of air blower **11** with a flashback preventer **111b**.

(104) To facilitate a further understanding of inventions of the present disclosure, the following description of FIG. 8 further teaches basic inventive principles of a thermal oxidation system and a thermal oxidization method of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various embodiments of thermal oxidation systems and thermal oxidization methods of the present disclosure.

(105) Referring to FIG. 8, a thermal oxidization system of the present disclosure incorporates a thermal oxidizer **50b** employing oxidation mixer **51**, an oxidation chamber **52a**, a retention chamber **53b** and a heat dissipater **54b** forming a fluid flow path for an oxidation of a waste gas stream **32** (e.g., TRI gases) into oxidized combustion products **74** (e.g., oxidized gases **73** (FIG. 2) including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2).

(106) As previously described, oxidation mixer **51** is structurally configured for mixing an oxidant **10** supplied by oxidant supply **11** (e.g., a blower or a pump) via a supply line **12** and a waste gas stream **32** (e.g., TRI gases) supplied by waste gas source **30** as controlled via a control flow conduit **33** (e.g., a solenoid valve and a flame arrestor in series coupling waste gas source **30** to oxidation mixer **51**) into a combustible waste gas stream **70**.

(107) In one embodiment, oxidation mixer **51** is a venturi air-gas mixer whereby turbulent fluid flows of oxidant **10** and waste gas stream **32** into the venture air-gas mixer are controlled via oxidant supply **11** and control flow conduit **33** to ensure combustible waste gas stream **70** attains proportional concentrations of oxidant **10** and waste gas stream **32** within a flammable range (e.g., **11.5:1** oxidant **10** to waste gas stream **32**). Additionally, oxidation mixer **51** may be equipped with a nozzle (not shown) for regulating a feeding of combustible waste gas stream **70** into oxidation chamber **52b** whereby the nozzle may be structurally configured to generate more turbulence to combustible waste gas stream **70**.

(108) Oxidation chamber **52a** is structurally configured for implementing a primary combustion reaction **71** therein of combustible waste gas stream **70** via a controlled emission of heat waves **56a** by a heating element **55a**. In one embodiment, oxidation chamber **52a** is a refractory ceramic

cylinder and heating element **55a** is embedded within the walls of the refractory ceramic cylinder. (109) Optionally, oxidation chamber **52a** may further employ a spark igniter **66** for a controlled ignition of combustible waste gas stream **70** at a proximal opening of oxidation chamber **52a**. For this embodiment, if oxidization mixer **51** is equipped with a nozzle, then a distal tip of spark igniter **66** may be positioned within or adjacent to the flow of the combustible waste gas stream **70** out of the nozzle into oxidation chamber **52a**.

(110) Combustible waste gas stream **70a** may fully or partially combust within oxidation chamber **52a**. Oxidation chamber **52a** therefore includes a supplement air inlet **52b** for an additional flow of oxidant **10** via oxidant supply line **14** or via atmosphere to supplement the oxygenated waste gas stream **72** flowing into retention chamber **53b**.

(111) Retention chamber **53b** is structurally configured for implementing a secondary combustion reaction **76** therein oxygenated waste gas stream **72** via a controlled omission of heat waves **56b** by a heating element **55b**. In one embodiment, retention chamber **53b** is a refractory ceramic cylinder and heating element **55b** is embedded within the walls of the refractory ceramic cylinder integrated with oxidation chamber **53b** as shown. Retention chamber **53b** ensures a complete oxidation of oxygenated waste gas stream **72** into heated oxidized combustion products **74** (e.g., oxidized gases **73** (FIG. 2) including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2).

(112) Heat dissipater **54b** is structurally configured for implementing a heat exchange with heated oxidized combustion products **74** to vent cooled oxidized combustion products **74** into the atmosphere. In one embodiment, heat dissipater **54b** includes a nozzle **59** having a cylindrical shape whereby oxidized combustion products **74** exits heat dissipater **54b** along a length and circumference of heat exchanger **57** with cooling atmosphere air **75** being directed vertically past heat exchanger **57** to thereby extract heat from heat exchanger **57**.

(113) The thermal oxidization system of FIG. 8 further incorporates a control system employing an oxidation controller **61**, a data logger **63** (FIG. 3) and a data reporter **65** (FIG. 3) housed within a control box **60** (FIG. 3).

(114) Oxidation controller **61** is structurally configured for controlling an operation of thermal oxidizer **50b** in accordance with the thermal oxidization method of FIGS. 4A-4C as previously described herein. More particularly, heating elements **55a** and **55b** are controlled via a similar scheme as heating element **55** (FIG. 2).

(115) In practice, oxidation controller **61** may be omitted whereby the heating elements **55** of thermal oxidizer **50a** (FIG. 2) and thermal oxidizer **50b** (FIG. 8) may be operated via an on/off switch.

(116) Also in practice, oxidation controller **61** may implement a variation of the thermal oxidization method **90** (FIGS. 4A-4C) excluding the heat dissipater thermocouple **69** and involving the oxidation chamber thermocouple **67** and/or the retention chamber thermocouple **68**. For such an embodiment, a one loop or a two loop proportional-integral-derivative ("PID") control may be implemented by oxidation controller **61** involving a zero crossover regulation of the heating element(s) **55** via heating regulation signal HR.

(117) To facilitate a further understanding of the inventions of the present disclosure, the following description of FIGS. 9-10 teach a couple of embodiments of the thermal oxidation system of FIG. 8. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various additional embodiments of thermal oxidization systems of the present disclosure.

(118) Referring to FIG. 9, an embodiment **250** of thermal oxidizer **50b** (FIG. 8) employs:

(119) 1. an oxidation mixer nozzle **251** having an air inlet on a side wall and a bottom gas inlet;

(120) 2. oxidation chamber **52a** (FIG. 8) in the form an oxidation chamber **252** including a cylindrical insulation blanket and a cylindrical thermal mass defining an oxidation fluid path with a spiral heating element **255a** encircling the oxidation fluid path;

(121) 3. retention chamber **53b** (FIG. 8) in the form of a retention chamber **253** including an

insulation blanket and a thermal mass defining a retention fluid path with a spiral heating element **255b** encircling the oxidation fluid path;

(122) 4. heat dissipater **54b** (FIG. **10A**) in the form of a heat dissipater **254** having a rain cap **290** (FIG. **9**); and

(123) 5. mounting brackets **291a** and **291b**.

(124) Thermocouples may or may not be incorporated dependent upon the control scheme.

(125) For example, FIG. **10A** illustrates an atmospheric version **250a** of thermal oxidizer **250** (FIG. **9**) involving a non-force flow of atmospheric air into the thermal oxidizer **250a** and an on/off control of the heating elements. A heat exchanger **257** provides for heat exchange and back flash protection of the heat dissipater, and a back flash **292** protects a back flash of the oxidation mixer.

(126) By further example, FIG. **10B** illustrates a force-air version **250b** of thermal oxidizer **250** (FIG. **9**) incorporating an air blower **211**, a solenoid valve **233a**, a flash arrestor **233b**, an oxidation thermocouple **267** and a retention thermocouple **268**.

(127) Referring to FIGS. **1-10**, those having ordinary skill in the art of the present disclosure will appreciate various advantages of the inventions of the present disclosure including, but not limited to:

(128) 1. an active or passive control of an efficient thermal combustion of toxic release inventories gases ("TRI"), particularly in compliance with applicable environment regulations; and

(129) 2. data logging and data reporting of the active control of the thermal combustion of TRI.

(130) To further facilitate an understanding of inventions of the present disclosure, the following description of FIGS. **11-14** teach basic inventive principles of a gas fired based thermal oxidization systems and gas fired based thermal oxidization methods of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of gas fired based thermal oxidization systems and gas fired based thermal oxidization methods of the present disclosure.

(131) Referring to FIG. **11**, an exemplary embodiment of a gas fired thermal oxidization system of the present disclosure incorporates a thermal oxidizer **300** employing an oxidation mixer **310** in the form of a venturi air-gas mixer, an oxidization chamber **302**, a retention chamber **303** and a heat dissipater (not shown) forming a fluid flow path for oxidation of a waste gas stream (e.g., TRI gases) into oxidized combustion products (e.g., oxidized gases including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2). As will be subsequently described in more detail herein, a gas fire based thermal oxidization method of the present disclosure represented by a flowchart **400** (FIG. **12**) implements an operation of thermal oxidizer **300** involving an ignition phase as best shown in FIG. **13A** for heating up chambers **302**, **303** and a combustion phase as best shown in FIG. **13B** for generating oxidized combustion products.

(132) Oxidation mixer **310** is structurally configured for implementing an ignitable mixture of an oxidant **10** and natural gas **35** during the ignition phase of the thermal oxidization method. The oxidant **10** is supplied to oxidation mixer **310** by an oxidant supply (not shown) (e.g., a blower or a pump) as controlled via a control flow conduit **312** (e.g., a solenoid and a modulation valve in series coupling the oxidant supply to oxidation mixer **310**). The natural gas **35** is supplied to oxidation mixer **310** by a natural gas source and a flame arrestor (not shown) as controlled via a control flow conduit **314** (e.g., a solenoid valve and a modulation valve in series coupling the natural gas source to oxidation mixer **310**).

(133) Oxidation mixer **310** is further structurally configured for implementing a combustible mixture of oxidant **10** and waste gas **32** during the combustion phase of the thermal oxidization method. Again, oxidant **10** is supplied to oxidation mixer **310** by an oxidant supply (not shown) (e.g., a blower or a pump) as controlled via a control flow conduit **312** (e.g., a solenoid valve and a flame arrestor in series coupling the oxidant supply to oxidation mixer **310**). The waste gas **32** is supplied to oxidation mixer **310** by a waste gas source and a flame arrestor (not shown) as

controlled via a control flow conduit **315** (e.g., a solenoid valve and a modulation valve in series coupling the waste gas source to oxidation mixer **310**).

(134) In one exemplary embodiment, oxidation mixer **310** is a venturi air-gas mixer whereby turbulent fluid flows of oxidant **10** and waste gas **32** into the venturi air-gas mixer are controlled via control flow conduits **312** and **316** to ensure proportional concentrations of oxidant **10** and waste gas stream **32** within a flammable range (e.g., **11.5:1** oxidant **10** to waste gas stream **32**).

(135) Still referring to FIG. **11**, oxidation mixer **310** is equipped with a burner nozzle **311** having a mounted spark igniter for igniting the ignitable mixture of oxidant **10** and natural gas **35** via oxidation mixer **310** during the ignition phase of the thermal oxidization method.

(136) During the combustion phase of the thermal oxidization method, burner nozzle **311** is further structurally configured for generating turbulence to the combustible mixture of oxidant **10** and waste gas **32** (i.e., a combustible waste gas stream) via oxidation mixer **310** and for regulating a feeding of the combustible mixture of oxidant **10** and waste gas **32** into oxidation chamber **302**.

(137) Still referring to FIG. **11**, during the ignition phase of the thermal oxidization method, oxidation chamber **302** and retention chamber **303** are structurally configured to absorb a flame generated by oxidation mixer **310** and burner nozzle **311** for elevating a thermal mass temperature within the chambers **302**, **303** above a thermal mass combustion threshold (e.g., $>1450^{\circ}\text{ F.}$).

(138) During the combustion phase of the thermal oxidization method, oxidation chamber **302** is further structurally configured for implementing a primary combustion reaction **71a** therein of the combustible waste gas stream **10**, **32** into an oxygenated waste gas stream, and retention chamber **303** is structurally configured for implementing a retention time for a secondary combustion reaction **71b** of the oxygenated waste gas stream into heated oxidized combustion products (e.g., oxidized gases including CO.sub.2 , H.sub.2O , N.sub.2 and O.sub.2).

(139) In one exemplary embodiment, oxidation chamber **302** and retention chamber **303** are integrally formed as a refractory ceramic cylinder with a diffuser **301** separating the chambers **302**, **303**.

(140) Still referring to FIG. **11**, the heat dissipator (not shown) may be structurally configured for implementing a heat exchange of atmosphere air with heated oxidized combustion products to vent cooled oxidized combustion products into the atmosphere. Alternatively, the heat dissipator may be structurally configured for implementing a vaporizing of oxidized combustion products **74** for a subsequent venting of some of the products to atmosphere (e.g., oxidized gases N.sub.2 and O.sub.2) and for a subsequent capture and storage of other products (e.g., oxidized gas CO.sub.2).

(141) Still referring to FIG. **11**, the thermal oxidization system of the present disclosure further incorporates a control system employing an oxidation controller **360**, a data logger (not shown) and a data reporter (not shown).

(142) Oxidation controller **360** is structurally configured for controlling an operation of thermal oxidizer **300** as will be further described herein in connection with a description of flowchart **400** of FIG. **12**.

(143) In one exemplary embodiment, oxidation controller **360** is an application specific main board or an application specific integrated circuit for controlling a thermal oxidation application of various inventive principles of the present disclosure as subsequently described herein in connection with FIG. **12**. The structural configuration of oxidation controller **360** may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for controlling an execution of the thermal oxidation application.

(144) A non-limiting example of oxidation controller **360** is an all-in-one XL4 controller sold by Homer APG of Indianapolis, Indiana that is configured in accordance with the inventive principles of the present disclosure.

(145) The data logger is structurally configured for logging operational data transmitted by oxidation controller **360** to the data logger via a push or pull operation, or by a monitoring of specific data points of oxidation controller **360** by the data logger. The operational data includes data informative of an operational status of thermal oxidizer **300** in executing the oxidation of waste gas stream **32**.

(146) In one exemplary embodiment, the data logger is an application specific main board or an application specific integrated circuit for controlling a data logging application of the present disclosure. The structural configuration of the data logger may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for executing the data logging application.

(147) A non-limiting example of the data logger is an all-in-one XLE controller sold by Homer APG of Indianapolis, Indiana that is configured in accordance with the inventive principles of the present disclosure. Another non-limiting example of the data logger is as an application module configured within oxidation controller **360**.

(148) Also in practice, the data logger may be omitted and oxidation controller **360** may be configured for executing the data logging application of the present disclosure.

(149) The data reporter **65** is structurally configured for receiving reporting data from the data logger via a push or pull operation and transmitting the reporting data to a network (e.g., a cellular network). The reporting data is the operational data processed by the data logger into a reporting format associated with the data reporter.

(150) In one embodiment, the data reporter is an application specific modem for executing a data reporting application of the present disclosure.

(151) A non-limiting example of the data logger **66** is a SonicU™ modem sold by SonicU, LLC of Greenfield, Indiana that is configured in accordance with the inventive principles of the present disclosure.

(152) Referring back to FIG. **11**, oxidation controller **360** regulates a control of thermal oxidizer **300** in accordance with the thermal oxidization method of the present disclosure as represented by a flowchart **400** shown in FIG. **12**. In support of executing the thermal oxidization method, (a) oxidation controller **360** senses and controls the generation of the flame via a connection of a flame detector **304** within combustion chamber **302** at a signal input **361s** of oxidation controller **360**, (b) oxidation controller **360** senses and controls a thermal mass temperature of retention chamber **303** via a connection of a thermocouple **305** within retention chamber **303** at a signal input **362s**, (c) oxidation controller **360** controls an activation and a deactivation of the spark igniter of burner nozzle **311** via a signal output **363s**, (d) oxidation controller **360** controls a feeding rate of natural gas **35** into oxidation mixer **310** via a connection of a signal output **364s** to control flow conduit **312**, (e) oxidation controller **360** controls a feeding rate of waste gas **32** into oxidation mixer **310** via a connection of a signal output **365s** to control flow conduit **314**, and (f) oxidation controller **360** controls a feeding rate of oxidant **10** into oxidation mixer **310** via a connection of a signal output **366s** to control flow conduit **316**.

(153) Referring to FIGS. **11** and **12**, prior to be oxidation controller **360** being powered-on, control flow conduits **312**, **314**, **316** are closed. Upon oxidation controller **360** being powered-on, oxidation controller **360** initiates a thermocouple validity check of thermocouple **305** involving a comparison by oxidation controller **360** of a thermal mass temperature of retention chamber **303** as sensed by thermocouple **305** to a thermocouple validity threshold. In practice, to test the validity of the thermocouple **305**, the thermocouple validity threshold should be set at an anticipated maximum room temperature (e.g., 22° C.) or to an anticipated maximum local outdoor air temperature.

(154) For the thermocouple validity test, if the thermal mass temperature of retention chamber **303**

is more than the thermocouple validity threshold, then oxidation controller **360** is powered down for a maintenance inspection of thermocouple **305** and powered on again after a successful repair/replacement, if any, of the thermocouple **305**.

(155) Otherwise, if the thermal mass temperature of retention chamber **303** is equal to or less than the thermocouple validity threshold, then oxidation controller **360** initiates a thermal oxidation limit check involving a comparison by oxidation controller **360** of a thermal mass temperature of retention chamber **303** as sensed by thermocouple **305** to a thermal oxidation limit threshold.

(156) In practice, to ensure a safe heat activation of chambers **302,303**, the thermal oxidation limit threshold should represent a predefined maximum temperature of retention chamber **303** prior to the heat activation of thermal oxidizer **300** (e.g., 300° C.) for all thermocouple limit thresholds).

(157) For the thermal oxidation limit check, if the thermal mass temperature of retention chamber **303** is more than the thermal oxidation limit threshold, then oxidation controller **360** is powered down for a cooling period whereby oxidation controller **360** may be powered up again upon expiration of the cooling period to re-execute the thermal oxidation limit check.

(158) Otherwise, if the thermal mass temperature of retention chamber **303** is equal to or less than the thermal oxidation limit threshold, then oxidation controller **360** executes a stage **S402** of flowchart **400** encompassing a natural gas flame ignition involving (a) an activation of the spark igniter of burner nozzle **311** via signal output **363s**, (b) a controlled feeding rate of natural gas **35** into oxidation mixer **310** via signal output **364s**, and (c) a controlled feeding rate of oxidant **10** into oxidation mixer **310** via signal output **366s**.

(159) Referring to FIGS. **12** and **13A**, during stage **S402**, to reach a thermal combustion threshold (e.g., 1450° F.), oxidation controller **360** may implement a modulation of the natural gas flame ignition involving (a) an increase in the feeding rate of natural gas **35** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** decreases, (b) a decrease in the feeding rate of natural gas **35** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** increases, (c) an increase in the feeding rate of oxidant **10** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** increases and (d) a decrease in the feeding rate of oxidant **10** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** decreases.

(160) In practice of stage **S402**, any increase or decrease of the feeding rate of natural gas **35** into oxidation chamber **310** is inversely proportional to any increase or decrease of the thermal mass temperature of retention chamber **303**, and any increase or decrease of the feeding rate of natural gas **35** into oxidation chamber **310** is directly proportional to any increase or decrease of the thermal mass temperature of retention chamber **303**.

(161) Upon the thermal mass temperature of retention chamber **303** being more than a thermal mass combustion threshold as determined by oxidation controller **360** during a stage **S404** of flowchart **400**, oxidation controller **360** executes a stage **S406** of flowchart **400** encompassing a waste gas combustion involving (a) a deactivation of the spark igniter of burner nozzle **311** via signal output **363s**, (b) a reduced feeding rate of natural gas **35** into oxidation mixer **310** via signal output **364s** or a closing of control flow conduit **312** via signal output **364s**, (c) a controlled feeding rate of waste gas **32** into oxidation mixer **310** via signal output **365s**, and (d) a controlled feeding rate of oxidant **10** into oxidation mixer **310** via signal output **366s**.

(162) Referring to FIGS. **12** and **13B**, during stage **S406**, to maintain a constant thermal mass temperature of retention chamber **303** (e.g., 1450° F.), oxidation controller **360** may implement a modulation of the waste gas combustion involving (a) a closing of control flow conduit **314** into oxidation mixer **310** via output signal **365s** when the thermal mass temperature of retention chamber **303** is more than the constant thermal mass temperature of retention chamber **303** and (b) an increase in the feeding rate of oxidant **10** into oxidation mixer **310** via output signal **366s** when the thermal mass temperature of retention chamber **303** is more than the constant thermal mass temperature of retention chamber **303**.

(163) Also during stage **S406**, if the thermal mass temperature of retention chamber **303** becomes more than the thermal oxidation limit threshold as determined by oxidation controller **360** during a stage **S408** of flowchart **400**, the oxidation controller **360** executes a stage **S410** of flowchart **400** encompassing an oxidizer shut down involving (a) a continued deactivation of the spark igniter of burner nozzle **311** via signal output **363s**, (b) a closing of control flow conduit **312** into oxidation mixer **310** via output signal **364s**, (c) a closing of control flow conduit **314** into oxidation mixer **310** via output signal **365s**, and (d) a closing of control flow conduit **316** into oxidation mixer **310** via output signal **366s**.

(164) Otherwise, oxidation controller **360** will continue to execute stages **S402-S408** as a function of the thermal mass temperature of retention chamber **303** as previously described herein for stages **S402-S408**.

(165) Referring to FIG. **14**, a thermal oxidizer **300'** is a version of thermal oxidizer **300** (FIG. **11**) that includes a supplement air inlet **306** for an additional flow of oxidant **10** via control flow conduit **316** as shown or via atmosphere to supplement the oxygenated waste gas stream flowing into retention chamber **303**.

(166) To further facilitate an understanding of inventions of the present disclosure, the following description of FIGS. **15** and **16** teaches basic inventive principles of a greenhouse gas capture based thermal oxidization system of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of greenhouse gas capture based thermal oxidization systems of the present disclosure.

(167) Referring to FIG. **15**, an exemplary greenhouse gas capture based thermal oxidation system of the present disclosure employs an embodiment **500** of a thermal oxidizer as previously described in the present disclosure, such as, for example, thermal oxidizer **50a** of FIG. **2**, thermal oxidizer **50b** of FIG. **8**, thermal oxidizer **300a** of FIG. **11** and/or thermal oxidizer **300b** of FIG. **14**.

(168) In operation, in accordance with the inventive principles of the present disclosure as set forth herein, thermal oxidizer **500** mixes an oxidant (e.g., atmospheric air) and waste gas (e.g., TRI gases) to yield a combustible waste gas stream **501** that sequentially undergoes a primary combustion reaction **502** into an oxygenated waste gas stream (not shown), a secondary combustion reaction **503** of the oxygenated waste gas stream into oxidized combustion products **504** (e.g., oxidized gases including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2 as shown). A final operational phase of thermal oxidizer involves a heat dissipation of oxidized combustion products **504** (e.g., a heat dissipation of >1750° F. to =300° F.).

(169) Still referring to FIG. **15**, the exemplary greenhouse gas processor based thermal oxidation system of the present disclosure further employs a greenhouse gas processor **700** in fluid communication with thermal oxidizer **500** (e.g., via a pipeline **600** as shown) to receive oxidized combustion products **504** at the reduced temperature. In one exemplary embodiment, thermal oxidizer **500** may include a heat exchanger to vent the oxidized combustion products **504** at the reduced temperature with the heat exchanger being enclosed by a sealed duct connector or the like to enable an unregulated or a regulated flow of the oxidized combustion products **504** at the reduced temperature to greenhouse gas processor **700**. In a second exemplary embodiment, thermal oxidizer **500** may include a vaporizer integrated with a pipeline or the like to enable an unregulated or a regulated flow of a water vaporization of the oxidized combustion products **504** to greenhouse gas processor **700** as will be further described herein.

(170) For purposes of describing and claiming the inventions of the present disclosure, greenhouse gas processor **700** broadly encompasses any machine in accordance with embodiments described in the present disclosure, as known in the art of the present disclosure and hereinafter conceived operable to extract greenhouse gas(es) **701** (e.g., oxidized gas CO.sub.2 as shown) of oxidized combustion products **504** from non-greenhouse gas(es) **702** (e.g., oxidized gases N.sub.2 and O.sub.2 as shown). Thereafter, greenhouse gas processor **700** may vent non-greenhouse gas(es) **702**

to atmosphere via a pipeline **601**, and may control a transportation and/or a storage of greenhouse gas(es) **701** via a pipeline **602**.

(171) In one exemplary embodiment, greenhouse gas processor **700** may water saturate the oxidized combustion products into a liquid condensate to extract the greenhouse gas(es) **701** within oxidized combustion products **504** from non-greenhouse gas(es) **702** via a greenhouse gas condensation process in accordance with embodiments of the present disclosure, as known in the art of the present disclosure or hereinafter conceived. For example, the greenhouse gas condensation process may involve a fog spraying of water onto the water vaporization of oxidized combustion products **504** to yield a liquid condensation of the greenhouse gas(es) **702** followed by an acid neutralization of the liquid condensation of the greenhouse gas(es) **702** as will be further described herein.

(172) Referring to FIG. **16**, an exemplary embodiment of the greenhouse gas capture based thermal oxidation system of FIG. **15** employs an embodiment **500a** of the thermal oxidizer **500** including an oxidation mixer **510**, an oxidation chamber **520**, a retention chamber **530**, and a heat dissipator in the form of a vaporizer **540**. In practice, vaporizer **540** broadly encompasses any device in accordance with embodiments of the present disclosure, as known in the art of the present disclosure and hereinafter conceived operable to utilizing a vaporizing agent (e.g., water) to facilitate a vaporization **504a** of the oxidized combustion products **504** flowing from retention chamber **530** to vaporizer **540** to thereby effect a heat dissipation of oxidized combustion products **504** (e.g., a reduction of heated oxidized gases from 1750° F.-1800° F. to ≈300° F.).

(173) In operation, in accordance with the inventive principles of the present disclosure as set forth herein, oxidation mixer **510** mixes an oxidant (e.g., atmospheric air) and waste gas (e.g., TRI gases) to yield a combustible waste gas stream (not shown), a primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream (not shown) occurs in oxidation chamber **520**, and a secondary combustion reaction of the oxygenated waste gas stream into heated oxidized combustion products occurs in retention chamber **530**.

(174) For this embodiment, vaporizer **540** is in fluid communication with a water supply **560** via a water isolation valve **603** and a water control valve **604** to fog spray of water onto the oxidized combustion products of retention chamber **530** flowing into vaporizer **540** to yield a water vaporization **504a** of oxidized combustion products **504** consisting of water vapor mixed with oxidized combustion products **504** at a reduced temperature. For example, a temperature of oxidized combustion products **504** exiting retention chamber **530** into vaporizer **540** may be approximately 1750° F. and vaporizer **540** may cool the oxidized combustion products **540** via the vaporization to a temperature of 300° F.

(175) Still referring to FIG. **16**, the exemplary embodiment of the greenhouse gas capture based thermal oxidation system of FIG. **15** further employs an embodiment **700a** of the greenhouse gas processor **700** including a condenser **710** and an acid neutralizer **720**.

(176) In practice, condenser **710** broadly encompasses any device in accordance with the present disclosure, as known in the art of the present disclosure or hereinafter conceived, that is operable to water saturate the water vaporization of the oxidized combustion products to a dew point temperature, and an acid neutralizer **720** broadly encompasses any device in accordance with the present disclosure, as known in the art of the present disclosure or hereinafter conceived for neutralizing acidic liquid.

(177) In operation, condenser **710** receives the water vaporization **504a** of oxidized combustion products **504** and is in fluid communication with water supply **560** via a water isolation valve **605** and a water control valve **606** to fog spray water onto the water vaporization **504a** of oxidized combustion products **504** to yield a saturation **504b** of oxidized combustion products **504** at a further reduced temperature. For example, condenser **710** may water saturate **504b** the vaporization **504a** of oxidized combustion products **504** to below a dew point temperature of 80° F. to yield an acidic greenhouse condensate (e.g., carbonic acid H.sub.2CO.sub.3 droplets as shown) and non-

greenhouse gas vapors (e.g., N.sub.2 and O.sub.2 as shown).

(178) Acid neutralizer **720** contains a neutralizing agent (e.g., a granular organic neutralizing agent like limestone, not shown) to facilitate neutralizing an acidic greenhouse condensate **701a** (e.g., neutral carbonic acid H.sub.2CO.sub.3). Acid neutralizer **720** thereafter separates the non-greenhouse gas(es) **702a** (e.g., oxidized gases N.sub.2 and O.sub.2 as shown) from neutralized acidic greenhouse condensate **701a**, whereby the non-greenhouse gas(es) **702a** may be vented to atmosphere via pipeline **601** and a transportation and/or a storage of substantially neutralized acidic greenhouse condensate **701a** may be controlled via pipeline **602**.

(179) To further facilitate an understanding of inventions of the present disclosure, the following description of FIGS. **17-19** teaches basic inventive principles of a greenhouse gas capture based thermal oxidization method of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of greenhouse gas capture based thermal oxidization methods of the present disclosure.

(180) Referring to FIG. **17**, a flowchart **800** represents an exemplary embodiment of a greenhouse gas capture based thermal oxidization method of the present disclosure. The following description of flowchart **800** will be in the context of a heat dissipating vaporizer **540a** as shown in FIG. **18**, a condenser **710a** as shown in FIG. **19**, and an acid neutralizer **720a** as shown in FIG. **19**.

(181) Referring to FIG. **18**, heat dissipating vaporizer **540a** employs tray type vaporizing chamber **541** having a breather cover **543**, a nozzle **542**, a pressure sensor **544** embedded into vaporizing chamber **541**, and a temperature sensor **546** embedded into vaporizing chamber **541**.

(182) Referring to FIG. **19**, condenser **710a** includes a condensing chamber **711**, a nozzle **712**, a vapor saturation chamber **713** and a temperature sensor **714** embedded into vapor saturation chamber **713**. Acid neutralizer **720a** includes a primary neutralizing chamber **722** enclosing a granular organic neutralizing material (e.g., limestone), a secondary neutralizing chamber **723** enclosing a granular organic neutralizing material (e.g., limestone), and a neutralized water chamber **724**.

(183) Referring to FIGS. **17** and **18**, a stage **S802** of flowchart **800** encompasses a water vaporization **504a** of oxidized combustion products **504** (e.g., oxidized gases including CO.sub.2, H.sub.2O, N.sub.2 and O.sub.2) exiting retention chamber **530** into vaporizing chamber **541**. More particularly, water supply **560** is controlled to supply a specified flow of water at a specified pressure to water control valve **604** (e.g., <1.0 gallons per hour at 60 psi). Temperature sensor **546** senses a temperature of vaporizing chamber **541** and a temperature controller **547** (e.g., a PID controller) controls water flow control valve **604** as a function of the sensed temperature of vaporizing chamber **541** to maintain a specified vaporization temperature point within vaporizing chamber **541** (e.g., 300° F.) as known in the art of the present disclosure. Water flowing through water flow control valve **604** is fogged sprayed via nozzle **542** onto the oxidized combustion products **504** flowing through the trays of vaporizing chamber **541** to yield a flow of the water vaporization **504a** of oxidized combustion products **504** at a reduced temperature through pipeline **600**. Pressure sensor **544** senses a pressure within vaporizing chamber **541** for purposes of controlling a flow rate of the water vaporization **504a** of oxidized combustion products **504** through pipeline **600** as will be subsequently described herein.

(184) Referring to FIGS. **17** and **19**, a stage **S804** of flowchart **800** encompasses a water saturation of vaporized oxidized combustion products **504a** flowing into condenser **710a**. More particularly, a motorized air pump **612** is operated via a motor speed controller **611** to pull the water vaporization **504a** of oxidized combustion products **504** through pipeline **600** into condensing chamber **711**. Motor speed controller **61** is controlled via a pressure controller **545** (e.g., a PID controller) to control the flow rate of the water vaporization **504a** of the oxidized combustion products **504** through pipeline **600** into condensing chamber **711** as a function of a pressure within vaporizing chamber **541** (FIG. **18**) as sensed by pressure sensor **544** as known in the art of the present

disclosure.

(185) Water supply **560** is controlled to supply a specified flow of water at a specified pressure to water control valve **606** (e.g., <1.0 gallons per hour at 60 psi), and water flowing through water flow control valve **606** is fogged sprayed via nozzle **712** upon vaporized oxidized combustion products **504a** flowing into fogging condensing chamber **711** to yield a water saturation of vaporized oxidized combustion products **504a** in vapor saturation chamber **713**. Temperature sensor **714** senses a temperature of vapor saturation chamber **713** and a temperature controller **715** (e.g., a PID controller) controls water flow control valve **606** as a function of the sensed temperature of vapor saturation chamber **713** to maintain a specified temperature point within condensation chamber **713** (e.g., 80° F.) as known in the art of the present disclosure. As a result, a condensation of greenhouse gas(es) drops through chamber **713** to the primary neutralizing chamber **722** of acid neutralizer **720a**.

(186) Still referring to FIGS. **17** and **19**, a stage **S806** of flowchart **800** encompasses an acid neutralization of the saturated oxidized combustion products. More particularly, the saturated oxidized combustion products downwardly percolate through the granular organic neutralizing material (e.g., limestone) of primary neutralizing chamber **722** whereby a substantially portion of the acidic greenhouse gas of the saturated oxidized combustion products dissolves into neutralized liquid **701a** maintained in a neutralized water chamber **724**, and the non-greenhouse gases of the saturated oxidized combustion products upwardly flowing through the granular organic neutralizing material (e.g., limestone) of secondary neutralizing chamber **723**. The neutralized liquid **701a** within neutralized water chamber **724** is continually or intermittently circulated via circulating pump **731** and a pipeline **732** to a nozzle **733** spraying the neutralized liquid **701a** into secondary neutralizing chamber **723**, whereby any residue of acidic greenhouse gas upwardly flowing through secondary neutralizing chamber **723** may also be dissolved into neutralized liquid **701a** maintained in neutralized water chamber **724**.

(187) Still referring to FIGS. **17** and **19**, a stage **S808** of flowchart **800** encompassing a venting of the non-greenhouse gases into atmosphere and a stage **S810** of flowchart **800** encompasses a storage of neutralized liquid **701a** for future transportation. More particularly, the motorized air pump **612** will pull the non-greenhouse gas(es) from secondary neutralizing chamber **723** through pipeline **602** and an exhaust **613** to atmosphere. Additionally, a control float switch **740** within a neutralized water chamber **724** is operated to control a water flow control valve **742** to drain neutralized liquid **701a** to a surge neutralized water tank **725** whenever a volume of neutralized liquid **701a** within neutralized water chamber **724** exceeds a specified volume. Additionally, a pressure equalizing valve **751** and a pressure isolation valve are operated to relieve pressure within neutralized water chamber **724** as a function of control float switch **740**.

(188) While various embodiments of the present invention have been illustrated and described, it will be understood by those skilled in the art that the methods and the system as described herein are illustrative, and various changes and modifications may be made and equivalents may be substituted for elements thereof without departing from the true scope of the present invention. In addition, many modifications may be made to adapt the teachings of the present invention without departing from its central scope. Therefore, it is intended that the present invention not be limited to the particular embodiments disclosed as the best mode contemplated for carrying out the present invention, but that the present invention include all embodiments falling within the scope of the appended claims.

Claims

1. A thermal oxidation system, comprising: a thermal oxidizer including an oxidation mixer, an oxidation chamber, a retention chamber and a heat dissipater forming a fluid flow path for a thermal oxidation of a waste gas, wherein the oxidation mixer is structurally configured to facilitate

a combustible mixture of a waste gas stream and an oxidant within the oxidation mixer into a combustible waste gas stream flowing within the oxidation mixer, wherein the oxidation chamber is in fluid communication with the oxidation mixer to receive a flow of the combustible waste gas stream, wherein the oxidation chamber includes a primary heating element or a gas burner to facilitate a primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream within the oxidation chamber, wherein the retention chamber is in fluid communication with the oxidation chamber to receive a flow of the oxygenated waste gas stream, wherein the retention chamber is structurally configured to facilitate a secondary combustion reaction of the oxygenated waste gas stream into oxidized gases within the retention chamber, wherein the oxidized gases includes at least one greenhouse gas, wherein the heat dissipater is in fluid communication with the retention chamber to receive a flow of the oxidized gases, wherein the heat dissipater is structurally configured to heat dissipate a flow of the oxidized gases within the heat dissipator; and a greenhouse gas processor in fluid communication with the heat dissipator to receive the oxidized gases flowing within the heat dissipator, wherein the greenhouse gas processor is structurally configured to extract the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor.

2. The thermal oxidization system of claim 1, wherein the heat dissipator structurally configured to heat dissipate the oxidized gases flowing within the heat dissipator includes: the heat dissipator being structurally configured to facilitate a water vaporization of the oxidized gases flowing within the heat dissipator.

3. The thermal oxidization system of claim 2, further comprising: a vaporization controller structurally configured to regulate the water vaporization by the heat dissipator of the oxidized gases flowing within the heat dissipator based on a temperature of the oxidized gases flowing within the heat dissipator.

4. The thermal oxidization system of claim 1, wherein the greenhouse gas processor structurally configured to extract the at least one greenhouse gas from the oxidized gases within the greenhouse processor includes: the greenhouse gas processor being structurally configured to form an acidic greenhouse gas condensate within the greenhouse gas processor.

5. The thermal oxidization system of claim 4, wherein the greenhouse gas processor structurally configured to liquid condensate the at least one greenhouse gas within the greenhouse gas processor includes: the greenhouse gas processor being structurally configured to facilitate a water saturation of the oxidized gases within the greenhouse gas processor.

6. The thermal oxidization system of claim 5, further comprising: a condensation controller structurally configured to regulate the water saturation by the condenser of the oxidized gases flowing within the condenser based on a temperature of the oxidized gases flowing within the condenser.

7. The thermal oxidization system of claim 1, wherein the greenhouse gas processor is further structurally configured to capture the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor.

8. The thermal oxidization system of claim 1, wherein the greenhouse gas processor is further structurally configured to form an acidic greenhouse gas condensate of the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor.

9. The thermal oxidization system of claim 8, wherein the greenhouse gas processor structurally configured to capture the liquid condensation of the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor includes: the greenhouse gas processor being structurally configured to acid neutralize the acidic greenhouse gas condensate by the greenhouse gas processor of the at least one greenhouse gas within the greenhouse gas processor.

10. The thermal oxidization system of claim 1, wherein the oxidized gases received by the greenhouse gas processor includes at least one non-greenhouse gas; and wherein the greenhouse gas processor is further structurally configured to vent the at least one non-greenhouse gas.

11. The thermal oxidization system of claim 1, further comprising: a thermocouple in one of: thermal communication with the oxidation chamber to measure a temperature of the oxidation chamber, or thermal communication with the retention chamber to measure a temperature of the retention chamber, or thermal communication with the heat dissipater to measure a temperature of the heat dissipater.
 12. The thermal oxidization system of claim 1, further comprising at least one of: a solenoid valve operable for regulating a feed of the waste gas stream into the oxidation mixer; and an oxidant supply operable for regulating a feed of the oxidant into the oxidation mixer.
 13. The thermal oxidization system of claim 1, wherein the retention chamber includes: a secondary heating element operable to facilitate a secondary combustion reaction of the combustible oxygenated waste gas stream into the oxidized gases within the retention chamber.
 14. The thermal oxidization system of claim 1, wherein the oxidation chamber further includes a supplemental air inlet structurally configured to facilitate a combustible mixture of an additional oxidant and the oxygenated waste gas stream flowing into the retention chamber.
 15. The thermal oxidization system of claim 1, further comprising: an oxidation controller structurally configured to regulate an operation of the thermal oxidizer including at least one of: regulating the combustible mixture of the waste gas stream and the oxidant into the combustible waste gas stream within the oxidation mixer; regulating the primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream within the oxidation chamber; and regulating the secondary combustion reaction of the oxygenated waste gas stream into the oxidized gases within the retention chamber.
 16. The thermal oxidization system of claim 15, further comprising: a thermocouple in thermal communication with the oxidation chamber to measure a temperature of the oxidation chamber; and wherein, responsive to a measurement of the temperature of the oxidation chamber by the thermocouple, the oxidation controller is structurally configured to monitor the temperature of the oxidation chamber relative to at least one regulation threshold representative of a controlled operation of the thermal oxidizer.
 17. The thermal oxidization system of claim 15, further comprising: a thermocouple in thermal communication with the retention chamber to measure a temperature of the retention chamber; and wherein, responsive to a measurement of the temperature of the retention chamber by the thermocouple, the oxidation controller is structurally configured to monitor the temperature of the retention chamber relative to at least one regulation threshold representative of a controlled operation of the thermal oxidizer.
 18. The thermal oxidization system of claim 15, further comprising: a thermocouple in thermal communication with the heat dissipater to measure a temperature of the heat dissipater; and wherein, responsive to a measurement of the temperature of the heat dissipater by the thermocouple, the oxidation controller is structurally configured to monitor the temperature of the heat dissipater relative to at least one regulation threshold representative of a controlled operation of thermal oxidizer.
 19. The thermal oxidization system of claim 15, further comprising at least one of: a data logger structurally configured to log a regulation of the operation of the thermal oxidizer by the oxidation controller; and a data reporter structurally configured to remotely report the regulation of the operation of the thermal oxidizer by the oxidation controller.
 20. The thermal oxidization system of claim 1, wherein the oxidation mixer is further structurally configured to facilitate an ignitable mixture of a natural gas stream and the oxidant within the oxidation mixer into an ignitable natural gas stream flowing with the oxidation mixer; wherein a gas burner is in fluid communication with the oxidation mixer to receive a flow of the ignitable nature gas stream within the oxidation mixer; and wherein the gas burner is structurally configured to ignite the ignitable nature gas stream within the oxidation chamber.
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